

Chapter 3: Numerical Sequences and Series

Section: Series of Nonnegative Terms

Based on Walter Rudin's *Principles of Mathematical Analysis*



Outline

Motivation

The Geometric Series

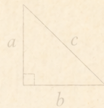
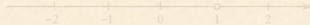
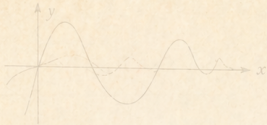
Cauchy Condensation Test

The p -Series

Logarithmic Series

The Elusive Boundary

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Why Nonnegative Terms?

- Comparison test (Theorem 3.25) needs **reference series** we trust



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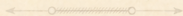
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- Goal: nail down **four reference series** as benchmarks

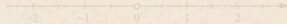


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The Geometric Series



$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



The Simplest Series of All

$$1 + x + x^2 + x^3 + x^4 + \dots$$

Question: When does this infinite sum make sense, and what is its value?

Idea: the partial sum $1 + x + \dots + x^n$ has a closed form.

Theorem 3.26: The Geometric Series

Theorem 3.26

If $0 \leq x < 1$, then

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

If $x \geq 1$, the series diverges.

- Convergent for $0 \leq x < 1$, with explicit sum
- Divergent for $x \geq 1$

Proof Strategy

Goal: understand $s_n = \sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$.

Plan:

1. Find a **closed form** for s_n when $x \neq 1$.
2. Take $n \rightarrow \infty$ to read off the sum.
3. Treat $x = 1$ separately by inspection.

Proof — Step 1: A Closed Form

Closed form for the partial sum

For $x \neq 1$,

$$S_n = \sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x}.$$

Check: $(1 - x)(1 + x + x^2 + \cdots + x^n) = 1 - x^{n+1}$ (telescoping).

Proof — Step 2: Let $n \rightarrow \infty$

Case $0 \leq x < 1$

Then $x^{n+1} \rightarrow 0$ as $n \rightarrow \infty$, so

$$s_n = \frac{1 - x^{n+1}}{1 - x} \longrightarrow \frac{1}{1 - x}.$$

Case $x > 1$

Then $x^{n+1} \rightarrow \infty$, so $s_n \rightarrow \infty$ and the series diverges.

Proof — Step 3: The Case $x = 1$

Case $x = 1$

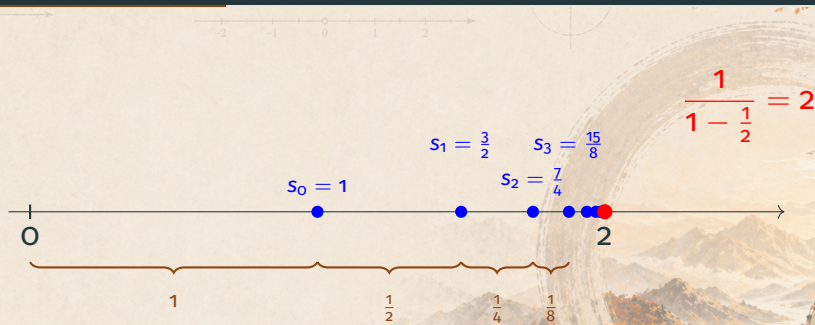
The series becomes

$$1 + 1 + 1 + 1 + \cdots ,$$

so $s_n = n + 1 \rightarrow \infty$. **Divergent.** □

Every case covered: convergent for $0 \leq x < 1$, divergent for $x \geq 1$.

Geometric Series: $x = \frac{1}{2}$



Each term covers *half* of the remaining gap to 2.

Two Quick Applications

Geometric series in action

- $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1}{1 - 1/2} = 2$

- $\sum_{n=0}^{\infty} \frac{1}{3^n} = \frac{1}{1 - 1/3} = \frac{3}{2}$

Why we care: the geometric series is the universal yardstick for the comparison and root/ratio tests to come.

Cauchy Condensation Test



A Stronger Tool for Monotone Series

- Suppose the terms **decrease monotonically**:
 $a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$.
- In applications, this is the *typical* situation.

Question: can we test convergence using only a few sample terms?

Theorem 3.27: Cauchy Condensation Test

Theorem 3.27

Suppose $a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + 16a_{16} + \cdots \quad (7)$$

converges.

Striking feature: a **very thin** subsequence determines convergence.

Intuition: Grouping by Powers of 2

Upper bound — blocks *starting* at 2^k , each term $\leq$ first term:

$$a_1 + \underbrace{(a_2 + a_3)}_{\leq 2a_2} + \underbrace{(a_4 + \cdots + a_7)}_{\leq 4a_4} + \underbrace{(a_8 + \cdots + a_{15})}_{\leq 8a_8} + \cdots$$

Lower bound — blocks *ending* at 2^k , each term $\geq$ last term:

$$a_1 + a_2 + \underbrace{(a_3 + a_4)}_{\geq 2a_4} + \underbrace{(a_5 + \cdots + a_8)}_{\geq 4a_8} + \underbrace{(a_9 + \cdots + a_{16})}_{\geq 8a_{16}} + \cdots$$

Both groupings reproduce $a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots$ up to a factor of 2.

Proof — Setup

Two partial sums

$$s_n = a_1 + a_2 + \cdots + a_n,$$

$$t_k = a_1 + 2a_2 + 4a_4 + \cdots + 2^k a_{2^k}.$$

By Theorem 3.24, both series converge iff their partial sums are bounded.

Strategy: show $\{s_n\}$ bounded $\iff \{t_k\}$ bounded.

$$f(x) = \int_a^x f(t) dt$$

Proof — Step 1: $s_n \leq t_k$

Upper bound: $n < 2^{k+1}$

Grouping the first n terms into blocks of length $1, 2, 4, \dots, 2^k$:

$$\begin{aligned} s_n &\leq a_1 + (a_2 + a_3) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}-1}) \\ &\leq a_1 + 2a_2 + 4a_4 + \cdots + 2^k a_{2^k} = t_k. \end{aligned}$$

Hence

$$s_n \leq t_k. \tag{8}$$

Proof — Step 2: $2s_n \geq t_k$

Lower bound: $n > 2^k$

Grouping again, but starting one step later:

$$\begin{aligned} s_n &\geq a_1 + a_2 + (a_3 + a_4) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k}) \\ &\geq \frac{1}{2} a_1 + a_2 + 2a_4 + \cdots + 2^{k-1} a_{2^k} = \frac{1}{2} t_k. \end{aligned}$$

Hence

$$2s_n \geq t_k. \tag{9}$$

Proof — Conclusion

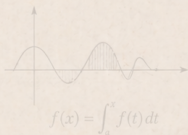
Combining (8) and (9)

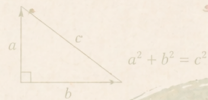
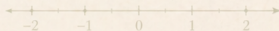
From inequalities (8) and (9):

$$s_n \leq t_k \quad \text{and} \quad t_k \leq 2s_n.$$

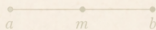
Therefore $\{s_n\}$ is bounded $\iff \{t_k\}$ is bounded.

By Theorem 3.24, the two series converge or diverge **together**. □





The p -Series



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Theorem 3.28: The p -Series Test

Theorem 3.28

The series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if $p > 1$ and diverges if $p \leq 1$.

Two famous corner cases:

- $p = 1$: the **harmonic series** $\sum 1/n$ **diverges**.
- $p = 2$: $\sum 1/n^2$ converges (the Basel problem: sum is $\pi^2/6$).

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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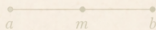
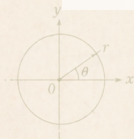
Proof — Step 1: The Easy Cases $p \leq 0$

Easy case: $p \leq 0$

If $p \leq 0$, then $n^{-p} \geq 1$ for all n , so $1/n^p \not\rightarrow 0$.

By Theorem 3.23, the series **diverges**.

So from now on, assume $p > 0$.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Proof — Step 2: Apply Cauchy Condensation

Condensation gives a geometric series

For $p > 0$, the sequence $\{1/n^p\}$ is positive and decreasing. Theorem 3.27 applies. The condensed series is

$$\sum_{k=0}^{\infty} 2^k \cdot \frac{1}{2^{kp}} = \sum_{k=0}^{\infty} 2^{(1-p)k}.$$

This is a **geometric series** with ratio $x = 2^{1-p}$.

Proof — Step 3: Compare with the Geometric Series

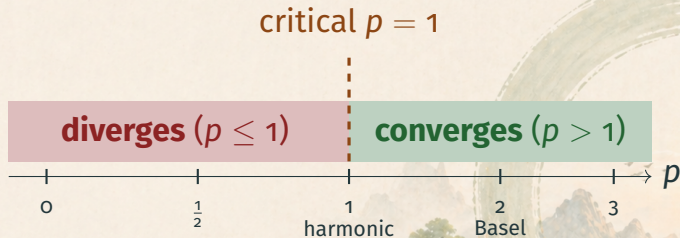
Reading off the answer from Theorem 3.26

By Theorem 3.26, the geometric series $\sum 2^{(1-p)k}$ converges iff

$$2^{1-p} < 1 \iff 1-p < 0 \iff p > 1.$$

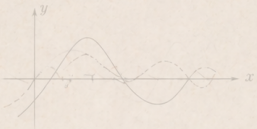
Combining with Step 1: $\sum 1/n^p$ converges $\iff p > 1$. □

The p -Series Landscape



The boundary $p = 1$ belongs to the divergent side.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

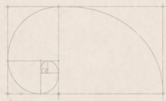


$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

Logarithmic Series



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

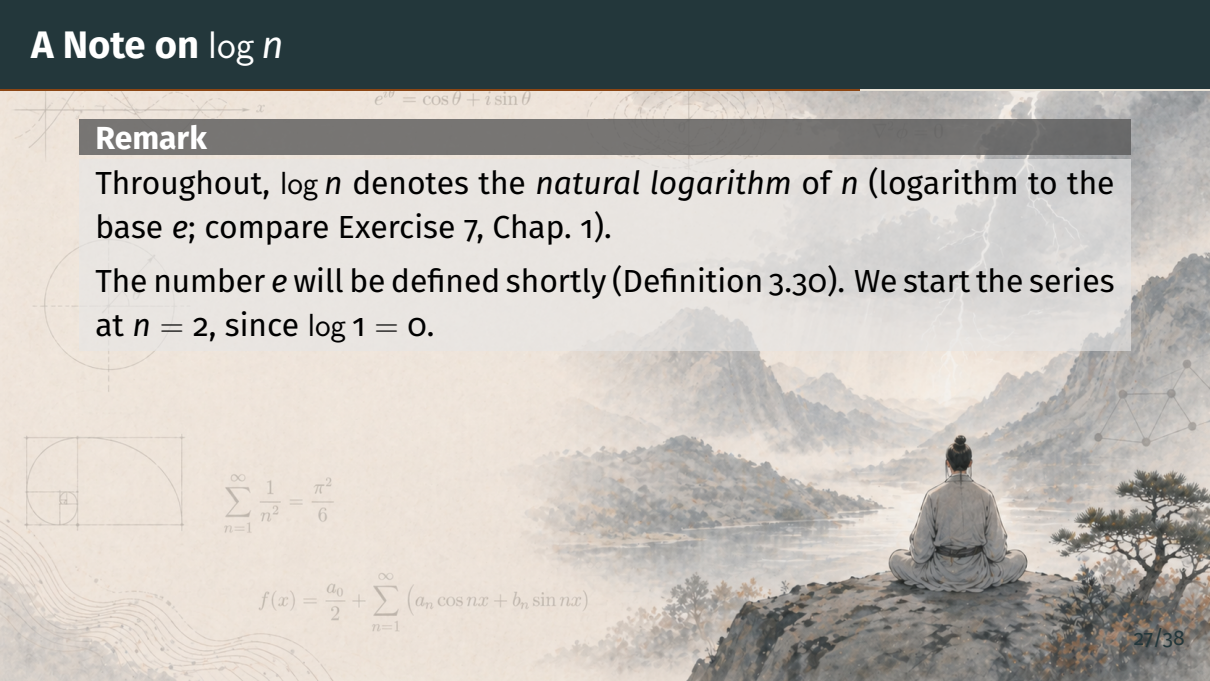


A Note on $\log n$

Remark

Throughout, $\log n$ denotes the *natural logarithm* of n (logarithm to the base e ; compare Exercise 7, Chap. 1).

The number e will be defined shortly (Definition 3.30). We start the series at $n = 2$, since $\log 1 = 0$.


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Theorem 3.29: Logarithmic p -Series

Theorem 3.29

If $p > 1$, the series

$$\sum_{n=2}^{\infty} \frac{1}{n (\log n)^p} \quad (10)$$

converges. If $p \leq 1$, the series diverges.

Same threshold $p = 1$ as in Theorem 3.28 — but with an extra log in the denominator.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.29

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Apply Cauchy condensation again

The logarithm is monotonically increasing, so $\{1/(n \log n)\}$ decreases. Theorem 3.27 applies. The condensed series is

$$\sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k (\log 2^k)^p} = \sum_{k=1}^{\infty} \frac{1}{(k \log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}. \quad (11)$$

A constant times a p -series. **Theorem 3.28** delivers the verdict. □

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

The Procedure Continues

Plugging more log factors into the denominator:

Example: equation (12)

$$\sum_{n=3}^{\infty} \frac{1}{n \log n \log \log n} \text{ diverges.}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

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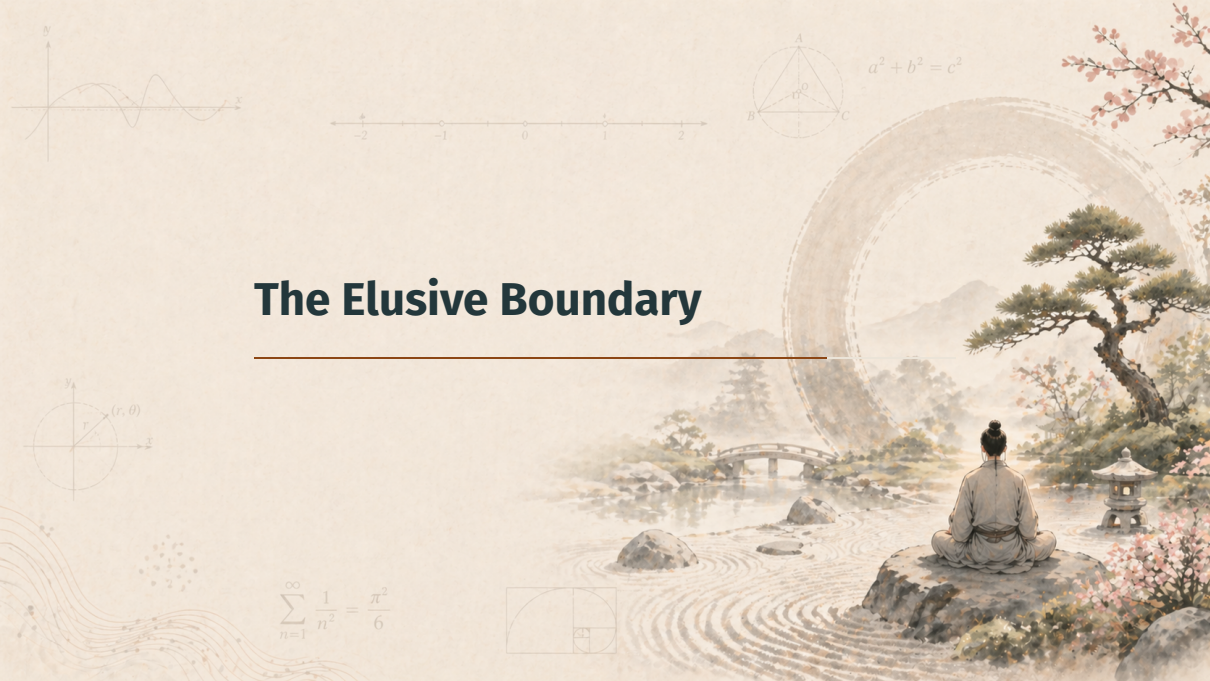
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Example: equation (13)

$$\sum_{n=3}^{\infty} \frac{1}{n \log n (\log \log n)^2} \quad \text{converges.}$$

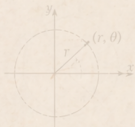
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



The Elusive Boundary

Is There a Sharp Boundary?

- Equations (12) and (13) differ only by an extra factor of $\log \log n$.
- One diverges, the other converges.

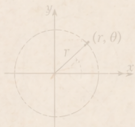


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



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- Equations (12) and (13) differ only by an extra factor of $\log \log n$.
- One diverges, the other converges.
- Iterating the procedure: even thinner gaps appear.



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



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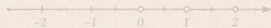
- Equations (12) and (13) differ only by an extra factor of $\log \log n$.
- One diverges, the other converges.
- Iterating the procedure: even thinner gaps appear.
- **No** such boundary exists — See Exercises 11(b), 12(b).

Knopp reference

See Knopp, *Theory and Application of Infinite Series*, Ch. IX, Sec. 41, for a fuller treatment.

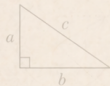
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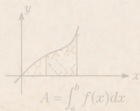


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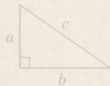
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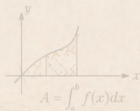
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Key takeaways:

- **Theorem 3.26 (geometric):** $\sum x^n \rightarrow 1/(1-x)$ iff $0 \leq x < 1$.



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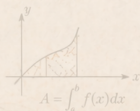
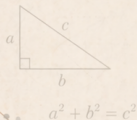


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- **Theorem 3.26 (geometric):** $\sum x^n \rightarrow 1/(1-x)$ iff $0 \leq x < 1$.
- **Theorem 3.27 (Cauchy condensation):** for $a_n \downarrow 0$, $\sum a_n$ and $\sum 2^k a_{2^k}$ converge (or diverge) together.

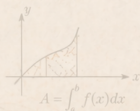
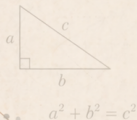


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- **Theorem 3.28 (p -series):** $\sum 1/n^p$ converges iff $p > 1$.



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- **Theorem 3.27 (Cauchy condensation):** for $a_n \downarrow 0$, $\sum a_n$ and $\sum 2^k a_{2^k}$ converge (or diverge) together.
- **Theorem 3.28 (p -series):** $\sum 1/n^p$ converges iff $p > 1$.
- **Theorem 3.29 (logarithmic):** $\sum 1/(n(\log n)^p)$ converges iff $p > 1$.

Coming up next: Definition of the number e and a careful estimate of its irrationality.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$